

Chapter 23

Properties of MLEs; Confidence Intervals

Lecture 23 | Devore 9e: 6.2, 7.1

23.1 More maximum likelihood

23.1.1 Several parameters at once

When there are m unknown parameters, set all m partial derivatives of the log-likelihood to zero and solve the system.

Example 23.1 (Normal, both parameters unknown). $X_1, \dots, X_n$ i.i.d. $\text{Normal}(\mu, \sigma^2)$. With $v = \sigma^2$,

$$\ell(\mu, v) = -\frac{n}{2} \log(2\pi v) - \frac{1}{2v} \sum_{i=1}^n (x_i - \mu)^2.$$

Then

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{v} \sum_i (x_i - \mu) = 0 \implies \hat{\mu} = \bar{x},$$

$$\frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{\sum_i (x_i - \mu)^2}{2v^2} = 0 \implies \hat{\sigma}^2 = \frac{1}{n} \sum_i (x_i - \hat{\mu})^2 = \frac{1}{n} \sum_i (x_i - \bar{x})^2.$$

So the MLE of σ^2 is the *biased* sample variance, $\frac{n-1}{n} S^2$.

23.1.2 Three properties worth knowing

Theorem 23.2 (Invariance). *If $\hat{\theta}$ is the MLE of θ and g is any function, then $g(\hat{\theta})$ is the MLE of $g(\theta)$.*

This is a genuine convenience: having found $\hat{\sigma}^2$, the MLE of σ is $\sqrt{\hat{\sigma}^2}$ — no new optimization. (Note that unbiasedness has no such property: $\mathbb{E}[\hat{\theta}] = \theta$ does *not* give $\mathbb{E}[g(\hat{\theta})] = g(\theta)$.)

Theorem 23.3 (Large-sample behavior). *Under mild regularity conditions, as $n \rightarrow \infty$ the MLE is*

- **consistent:** $\hat{\theta} \rightarrow \theta$;
- **asymptotically unbiased:** $\mathbb{E}[\hat{\theta}] \rightarrow \theta$;
- **asymptotically normal and efficient:** *no other estimator has smaller variance in the limit.*

Remark 23.4 (MLE versus method of moments).

	Method of moments	Maximum likelihood
effort	solve algebraic equations	maximize a function
uses	the first m moments only	the whole density
guarantees	few	consistent, asymptotically efficient
risk	can leave the parameter space	usually well behaved

Method of moments is a good way to get a starting value; maximum likelihood is what you report.

23.2 From point estimates to intervals

A point estimate is a single number and is therefore almost surely wrong. Reporting $\hat{\mu} = 8.3$ says nothing about whether the truth is plausibly 8.2 or plausibly 40. What we want is an interval, together with a statement of how often that interval works.

Definition 23.5 (Confidence interval). A $100(1 - \alpha)\%$ **confidence interval** for θ is a pair of statistics $L = L(X_1, \dots, X_n)$ and $U = U(X_1, \dots, X_n)$ with

$$\mathbb{P}\{L \leq \theta \leq U\} = 1 - \alpha.$$

23.2.1 The normal mean, σ known

Let $X_1, \dots, X_n$ be i.i.d. $\text{Normal}(\mu, \sigma^2)$ with σ known. From Lecture 19, $\bar{X} \sim \text{Normal}(\mu, \sigma^2/n)$ exactly, so

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \text{Normal}(0, 1).$$

Let $z_{\alpha/2}$ be the value with $\mathbb{P}\{Z > z_{\alpha/2}\} = \alpha/2$. By symmetry,

$$\mathbb{P}\left\{-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right\} = 1 - \alpha.$$

Now solve the inequality for μ — this is the only real step:

$$\boxed{\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}}$$

confidence	α	$z_{\alpha/2}$	interval
90%	0.10	1.645	$\bar{x} \pm 1.645 \sigma / \sqrt{n}$
95%	0.05	1.96	$\bar{x} \pm 1.96 \sigma / \sqrt{n}$
99%	0.01	2.576	$\bar{x} \pm 2.576 \sigma / \sqrt{n}$

Example 23.6. A machine fills bottles with $\sigma = 0.5$ ml. A sample of $n = 25$ bottles has $\bar{x} = 500.2$ ml. A 95% confidence interval for μ is

$$500.2 \pm 1.96 \cdot \frac{0.5}{\sqrt{25}} = 500.2 \pm 0.196 = [500.00, 500.40].$$

Remark 23.7 (What “95% confident” means — and does not). μ is a fixed unknown number; it is not random. The *interval* is random, because $\bar{X}$ is. The correct reading is:

if we repeated the whole experiment many times, about 95% of the intervals so constructed would contain μ .

It is **not** correct to say “ μ has a 95% chance of lying in $[500.00, 500.40]$ ” — once the data is in, either it does or it does not.

Remark 23.8 (Width). The width is $2z_{\alpha/2}\sigma/\sqrt{n}$. So:

- more confidence $\Rightarrow$ wider interval (you cannot get both);
- to halve the width, take 4 times as many observations — the $\sqrt{n}$ law again.

23.2.2 Large samples, σ unknown

In practice σ is unknown too. For large n , replace it by S and appeal to the CLT: for *any* underlying distribution with finite variance,

$$\mu \approx \bar{X} \pm z_{\alpha/2} \frac{S}{\sqrt{n}} \quad (n \text{ large}).$$

For small n from a normal population, $z_{\alpha/2}$ must be replaced by a t -distribution quantile — which is the starting point of STAT401.

Example 23.9 (Better than a coin?). A trading signal calls the direction of the market correctly on 60 of 100 days. Give a 95% confidence interval for its long-run hit rate p .

Each day is a Bernoulli variable, 1 for a correct call and 0 otherwise, so $\bar{X} = \hat{p} = 0.6$ and $\sigma^2 = p(1-p)$, which we estimate by $\hat{p}(1-\hat{p}) = 0.24$. The large-sample interval is

$$0.6 \pm 1.96 \sqrt{\frac{0.24}{100}} = 0.6 \pm 0.096 = [0.504, 0.696].$$

It just excludes 0.5, so the data are some evidence that the signal beats a coin.

Remark 23.10 (Many tries). Now suppose this signal was the best of 20 that were tried, each tested on its own data, and in truth all of them are no better than a coin. For each one, the interval contains the true 0.5 with probability 0.95. The chance that at least one of the 20 intervals excludes 0.5 anyway is

$$1 - 0.95^{20} \approx 0.64.$$

So a result like the one above is weak evidence if it is the best of many attempts. This is the *multiple testing* problem, and it is the first thing to ask about any impressive-looking backtest: how many things were tried before this one?

Remark 23.11. Everything in this course has been building to this sentence: *data* $\rightarrow$ *estimator* $\rightarrow$ *sampling distribution* $\rightarrow$ *interval with a guarantee*. The probability half of the course exists precisely to make the third arrow possible.